

Digital Image Processing

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Lecture Outline

- Digital Image Processing
- Image Representation
- Basic Image Operations and Transformations
- Image Histogram and Equalization
- Spatial Filtering

Digital Image Processing

Digital Image:

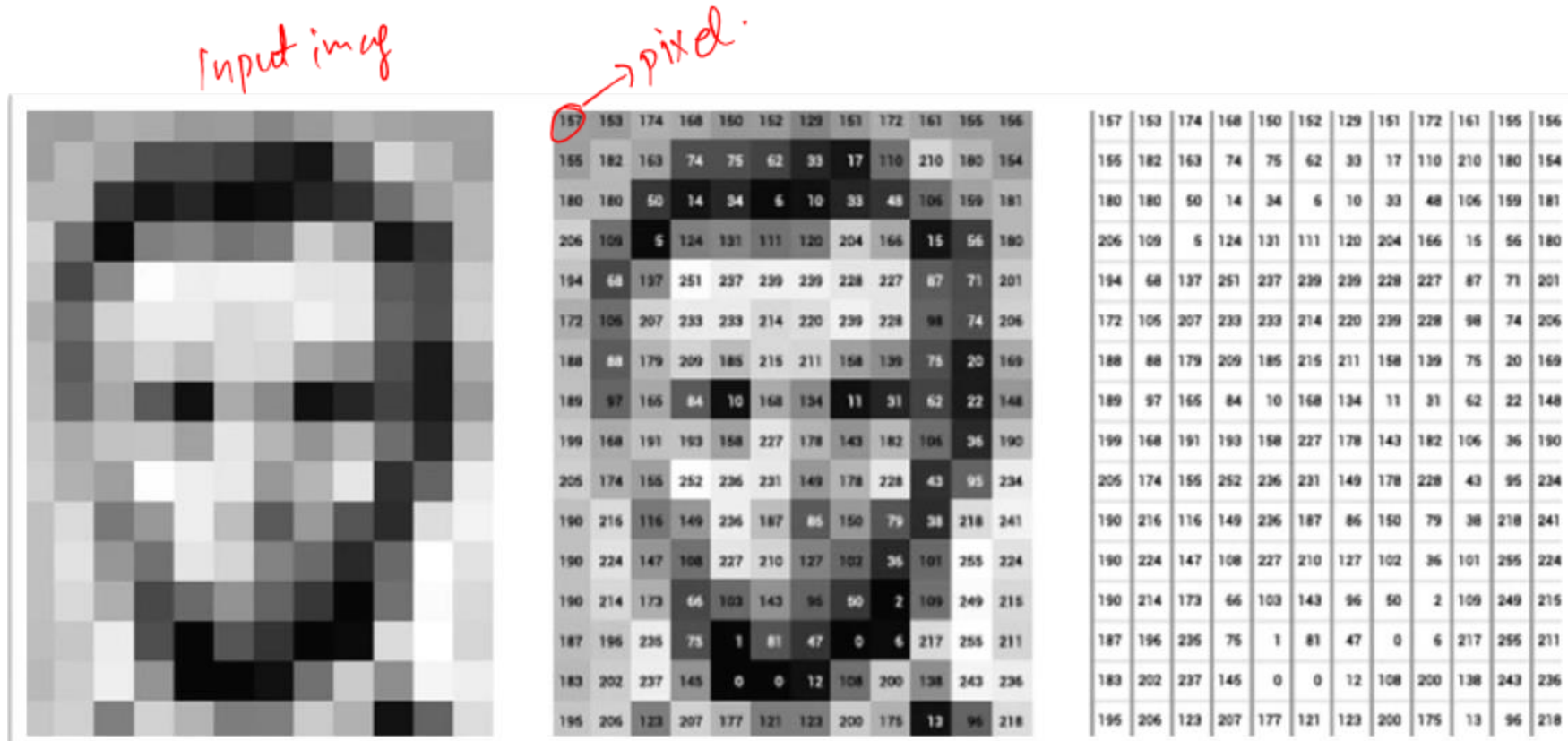
- Representation of two-dimensional image as finite set of digital values.

Digital Image Processing (DIP)

- Processing digital images using computational algorithms.
- Two main tasks of DIP:
 1. Improve image information for human's perception and interpretation;
 2. Process image for storage, transmission, and representation of autonomous machine perception.

Digital Image Processing (DIP)

- **Pixel:**
- The basic component/building block of an image

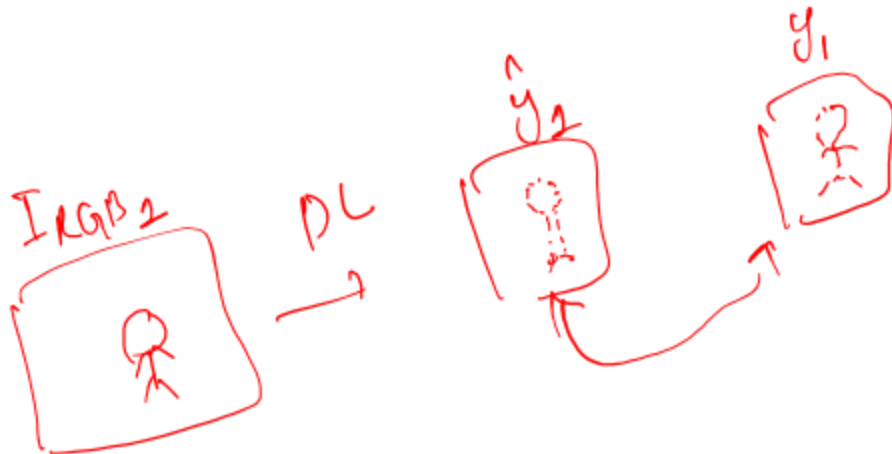


Digital Image Processing (DIP)

- **Image Similarity/Difference**
- Suppose different pixel values such as p, q, and z with varied coordinates (x, y), (s, t), and (u, v), respectively.
- The distance function D will satisfy three main properties:
 - **Non-negativity**

① $D(p, q) \geq 0$

$D(p, q) = 0, p = q$



Digital Image Processing (DIP)

- **Image Similarity/Difference**
- Suppose different pixel values such as p , q , and z with varied coordinates (x, y) , (s, t) , and (u, v) , respectively.
- The distance function D will satisfy three main properties:
 - Non-negativity
 - **Symmetry**

$$D(p, q) = D(q, p)$$



Digital Image Processing (DIP)

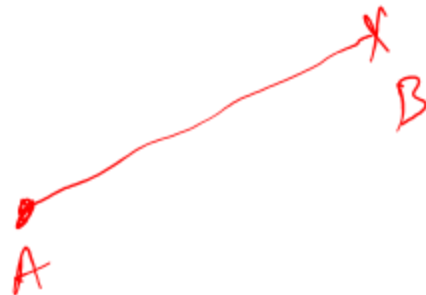
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- Suppose different pixel values such as p , q , and z with varied coordinates (x, y) , (s, t) , and (u, v) , respectively.
- The distance function D will satisfy three main properties:
 - Non-negativity
 - Symmetry
 - **Triangular inequality**

$$D(p, z) \leq D(p, q) + D(q, z)$$

Digital Image Processing (DIP)

- **Image Similarity/Difference**
- Basic types of distance measures:
 1. **Euclidean distance**

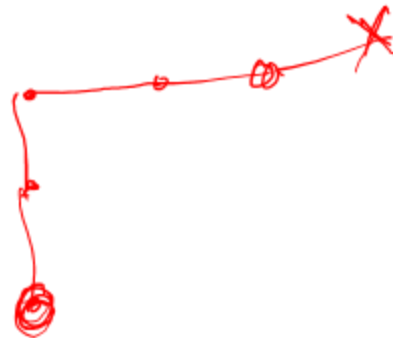
$$D(p, q) = \sqrt{(x - s)^2 + (y - t)^2}$$



Digital Image Processing (DIP)

- **Image Similarity/Difference**
- Basic types of distance measures:
 1. Euclidean distance
 2. **Manhattan distance (city block distance)**

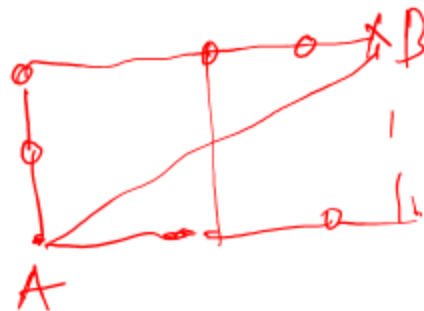
$$MD(P, q) = |x - s| + |y - t|$$



Digital Image Processing (DIP)

- **Image Similarity/Difference**
- Basic types of distance measures:
 1. Euclidean distance
 2. Manhattan distance (city block distance)
 3. **Chess board distance**

$$D(p, q) = \max(|x - s|, |y - t|)$$

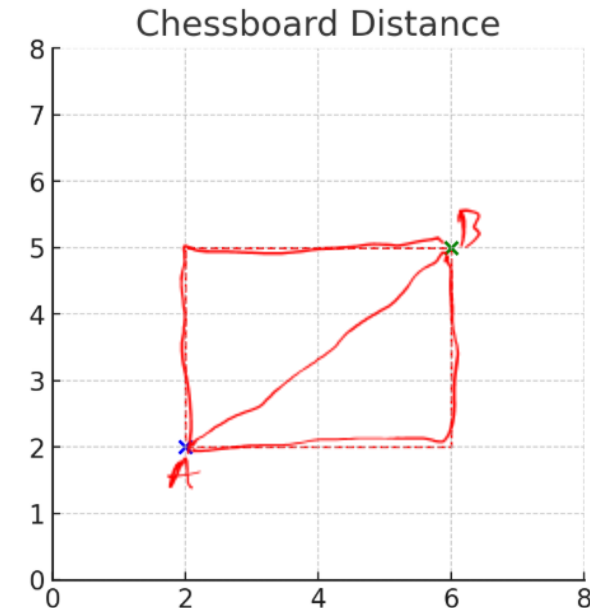
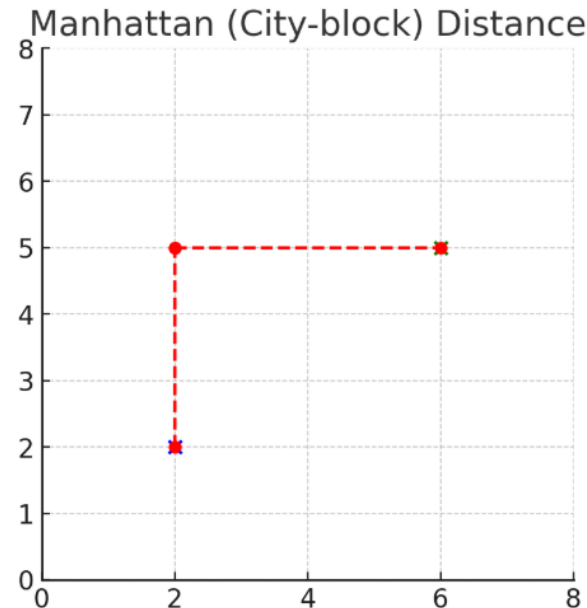
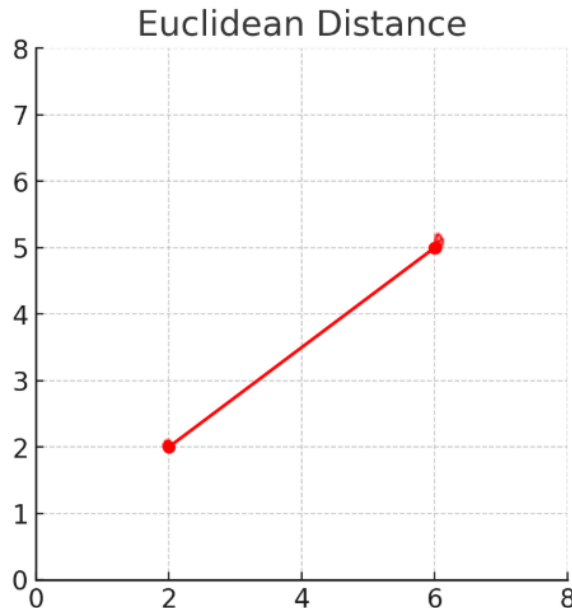


structure similarity index measurement.
(SSIM).



Digital Image Processing (DIP)

- Image Similarity/Difference
- Basic types of distance measures:
 1. Euclidean distance
 2. Manhattan distance (city block distance)
 3. Chess board distance



Digital Image Processing (DIP)

- Image Similarity/Difference
- Basic types of distance measures:

1. Euclidean distance?

2. Manhattan distance (city block distance)?

3. Chess board distance?

$p = (x, y) = (2, 2)$, $q = (x, y) = (6, 5)$
 $ED = \sqrt{(2 - 6)^2 + (2 - 5)^2}$
 $= \sqrt{16 + 9}$
 $= \sqrt{25} = 5$

$(1, 1)$	$1, 2$	$(1, 3)$	$(1, 4)$	$(1, 5)$
0	0	0	0	0
0	0	1	1	0
0	1	1	0	0
0	1	0	0	0
0	0	0	0	0
0	0	0	0	0

$$\sqrt{(2 - 4)^2 + (4 - 2)^2}$$

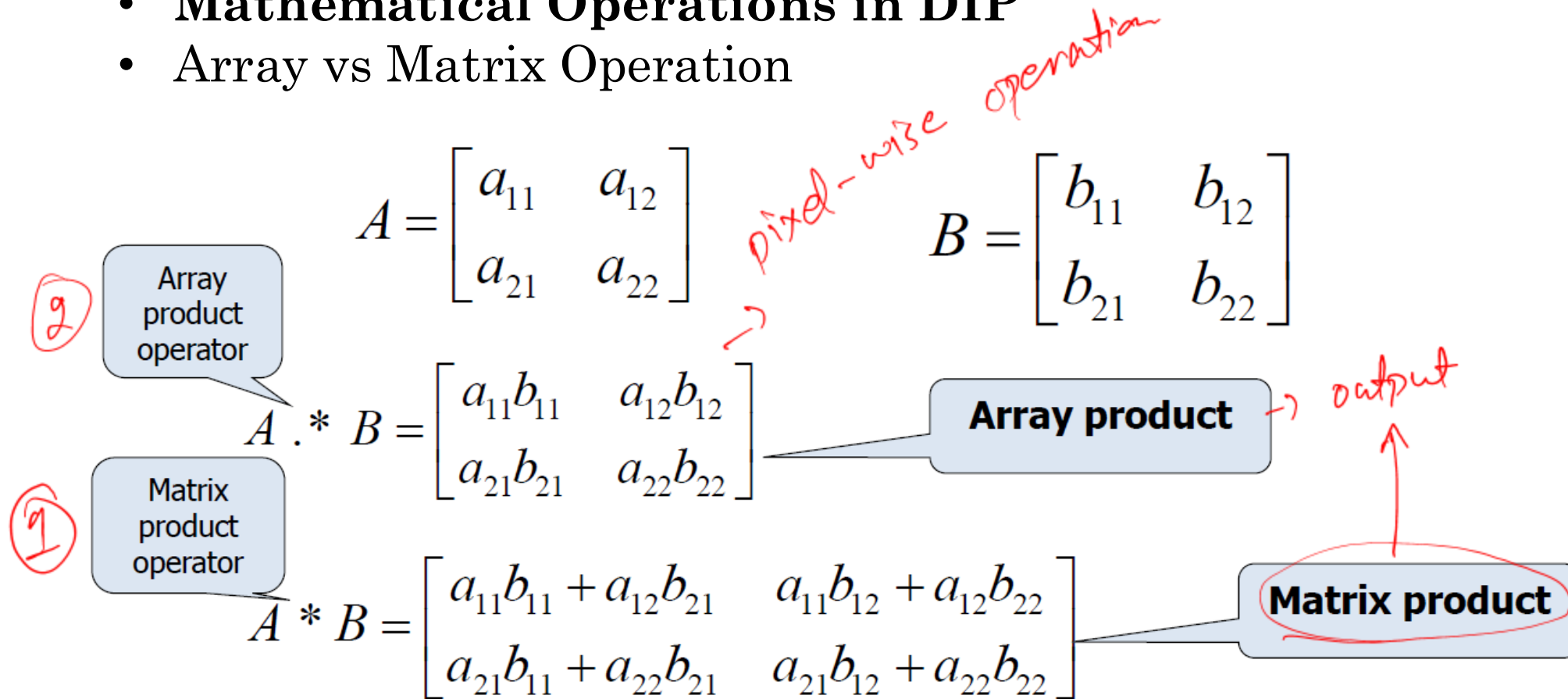
$$\sqrt{4 + 4}$$

$$= \sqrt{8}$$

$$\approx 2.8$$

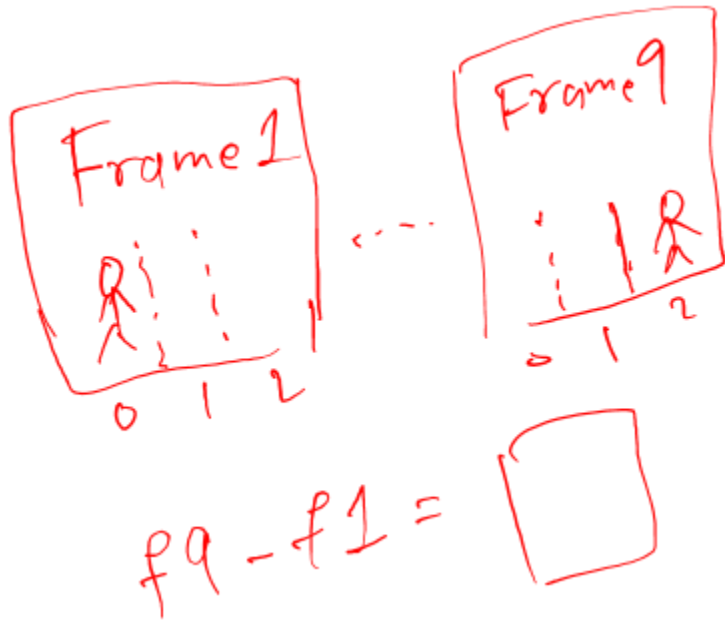
Digital Image Processing (DIP)

- Mathematical Operations in DIP
- Array vs Matrix Operation



Digital Image Processing (DIP)

- **Arithmetic Operations in DIP**
- These are array operations between images such as:



$$s(x,y) = f(x,y) + g(x,y)$$

$$d(x,y) = f(x,y) - g(x,y)$$

$$p(x,y) = f(x,y) \times g(x,y)$$

$$v(x,y) = f(x,y) \div g(x,y)$$



Digital Image Processing (DIP)

- **Arithmetic Operations in DIP**
- Example: Addition of noisy images for noise reduction

Input image $f(x, y)$
random noise $n(x, y)$

observed image $g(x, y) = f(x, y) + n(x, y)$

$\sum_{\text{Ave}} (n(x, y)) = 0$

$g_i(x, y) = f(x, y) + n_{i=1, \dots, K}(x, y)$

$\bar{g}(x, y) = \frac{1}{K} \sum_{i=1}^K g_i(x, y)$

$\bar{g}(x, y) = f(x, y) + \frac{1}{K} \sum_{i=1}^K n_i(x, y)$

constant

positive negative

Digital Image Processing (DIP)

- **Spatial Domain Process:**
- The two basic categories of spatial processing are intensity transformations and spatial filtering.
 1. **Intensity transformations** are applied to a single pixel of the image for contrast enhancement and image thresholding.
 2. **Spatial filtering** handles processes such as sharpening by processing in the neighborhood of each pixel in the image.

Digital Image Processing (DIP)

- **Intensity transformation**

$$g(x, y) = T[f(x, y)]$$

$f(x, y)$: input image

$g(x, y)$: output image

T : an operator on f defined over
a neighborhood of point (x, y)

Digital Image Processing (DIP)

- Intensity transformation

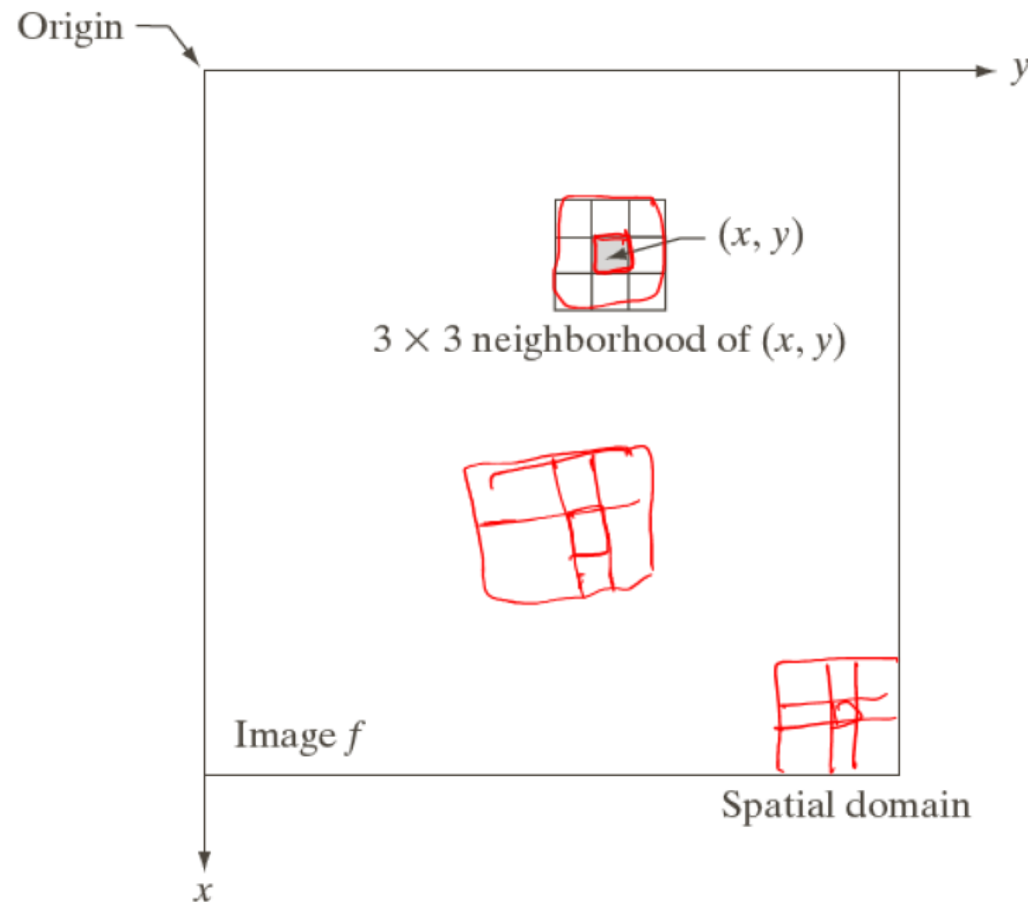


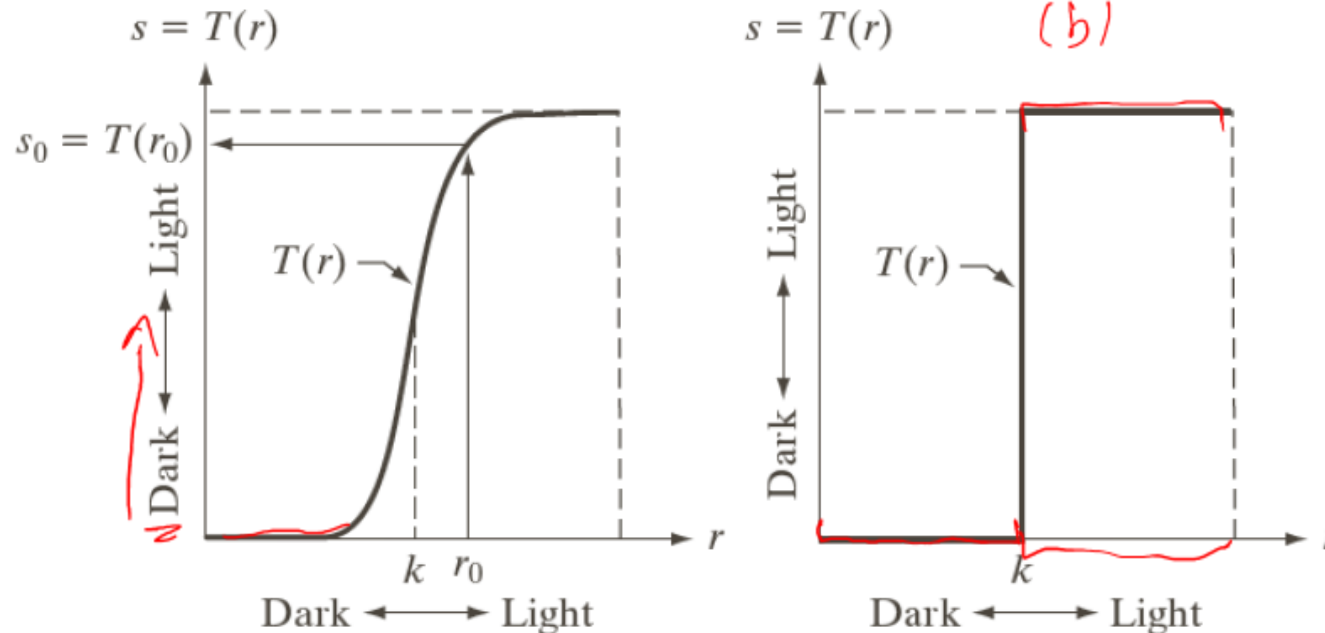
FIGURE 3.1

A 3×3 neighborhood about a point (x, y) in an image in the spatial domain. The neighborhood is moved from pixel to pixel in the image to generate an output image.

Digital Image Processing (DIP)

- Intensity transformation

$$s = T(r)$$



a b

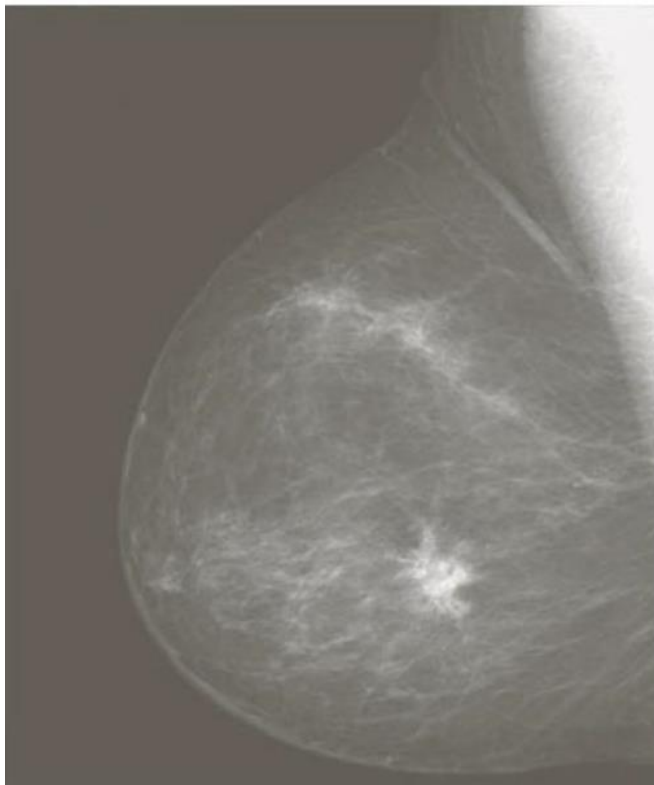
FIGURE 3.2
Intensity transformation functions.
(a) Contrast-stretching function.
(b) Thresholding function.

Digital Image Processing (DIP)

- Image Negatives

$$s = L - 1 - r$$

Handwritten annotations: $0-255$ (pointing to $L-1$) and *current pixel value* (pointing to r)



a b

FIGURE 3.4

(a) Original digital mammogram.

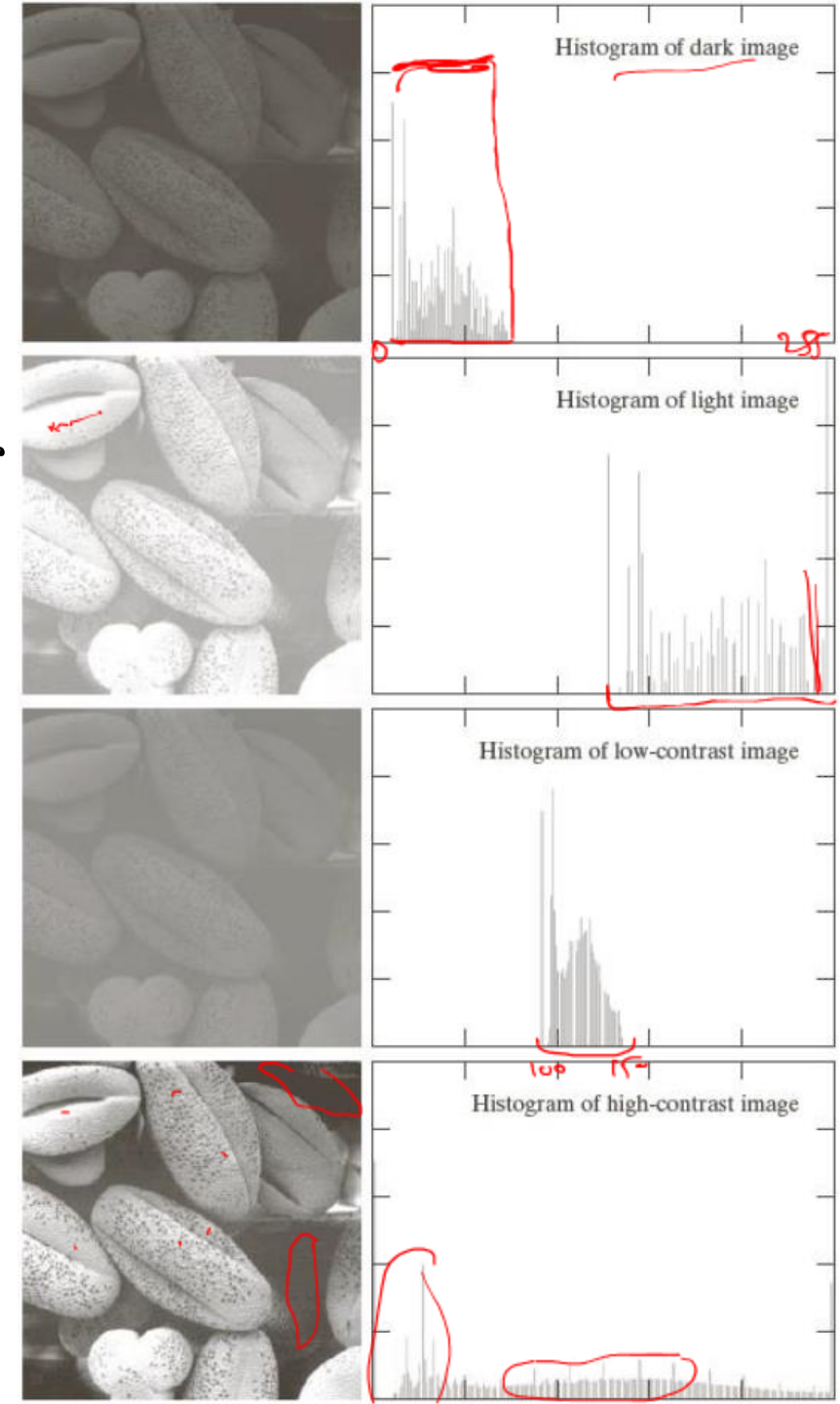
(b) Negative image obtained using the negative transformation in Eq. (3.2-1).

(Courtesy of G.E. Medical Systems.)

Small lesion

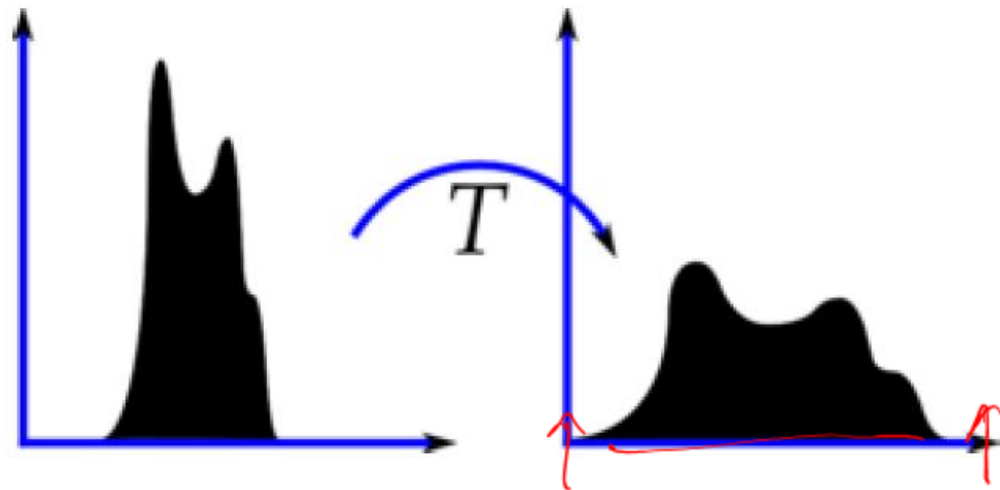
Digital Image Processing (DIP)

- **Image Histograms:**
- A graphical representation of the distribution of the gray values in an image.



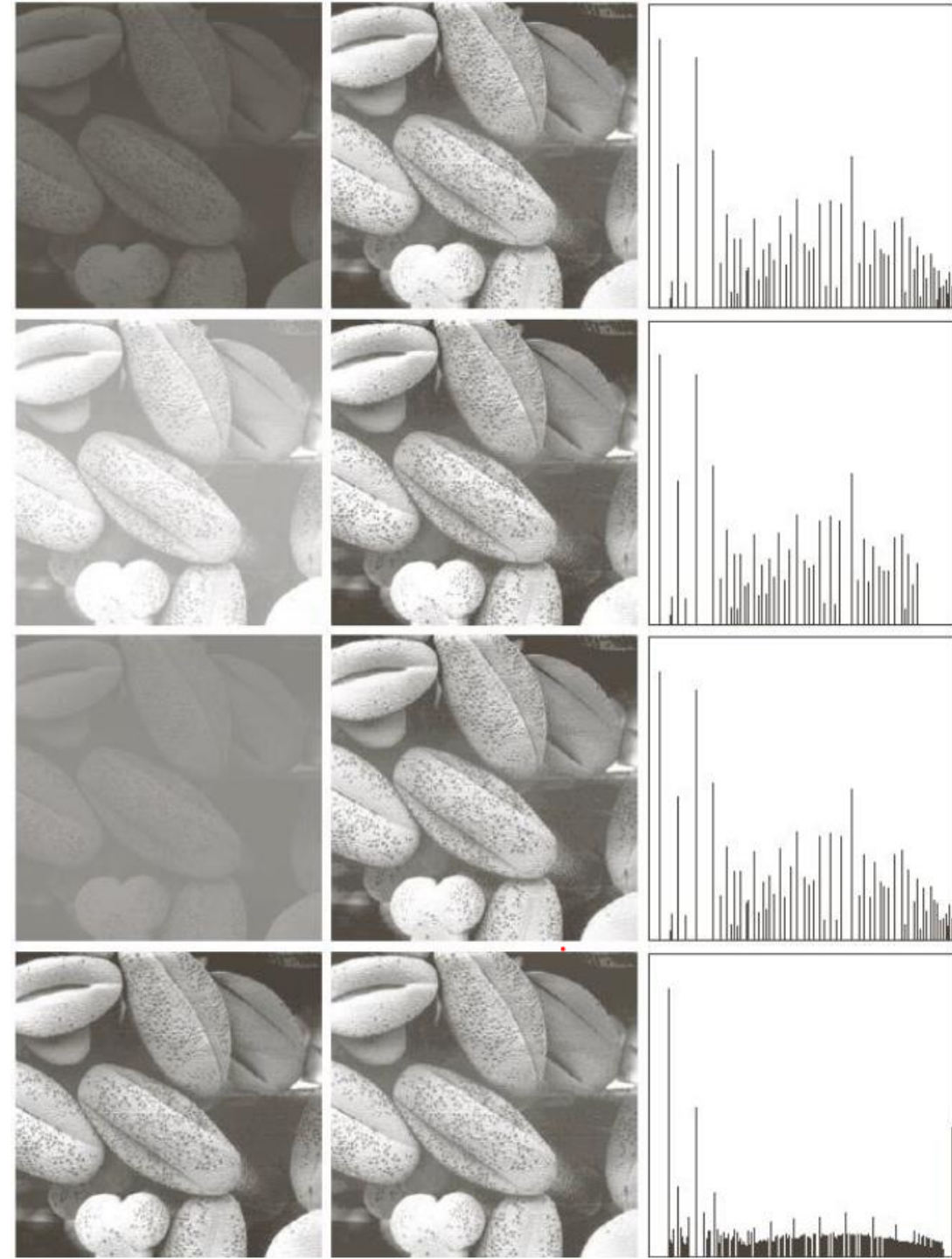
Digital Image Processing (DIP)

- **Histogram Equalization:**
- Consider an image whose pixel values are limited to only a certain range of values.
- For example, in a brighter image all pixels will be limited to higher values.
- But a good image should have pixels from all regions of the image.
- Therefore, stretch the histogram to both ends and that's what Histogram Equalization does, normally improves the contrast of the image.



Digital Image Processing (DIP)

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Digital Image Processing (DIP)

- **Spatial Filtering:**
- A spatial filter consists of a neighbourhood and predefined operation.
- **Neighbourhood:** a small region around the pixel (e.g., 3×3 , 5×5 window).

Digital Image Processing (DIP)

- **Spatial Filtering:**
- A spatial filter consists of a neighbourhood and predefined operation.
- Neighbourhood: a small region around the pixel (e.g., 3×3 , 5×5 window).
- **Predefined operation:** rule applied on that neighborhood (e.g., averaging, sharpening, edge detection).

Digital Image Processing (DIP)

- **Spatial Filtering: Moving Average in 2D**

$$F[x, y]$$

[illegible]

$$G[x, y]$$

A 10x10 grid with a red square at (1,1) containing the number 0.

Digital Image Processing (DIP)

- Spatial Filtering: Moving Average in 2D

$F[x, y]$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$G[x, y]$

	0	10							

sum
1 (90)
3x3
1 (90)
9

Digital Image Processing (DIP)

- Spatial Filtering: Moving Average in 2D

$$F[x, y]$$

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$$G[x, y]$$

	0	10	20						

$\frac{1}{9} \times 180$

Digital Image Processing (DIP)

- ~~Spatial Filtering: Moving Average in 2D~~

$$F[x, y]$$

[illegible]

$$G[x, y]$$

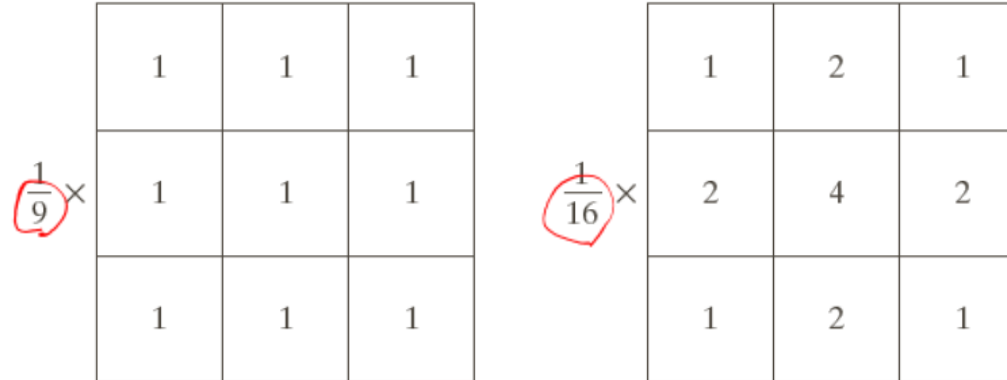
[illegible]

Digital Image Processing (DIP)

- **Spatial Filtering: Smoothing**
- Smoothing filters are used for blurring and for noise reduction.
- Blurring is used in removal of small details and bridging of small gaps in lines or curves

Digital Image Processing (DIP)

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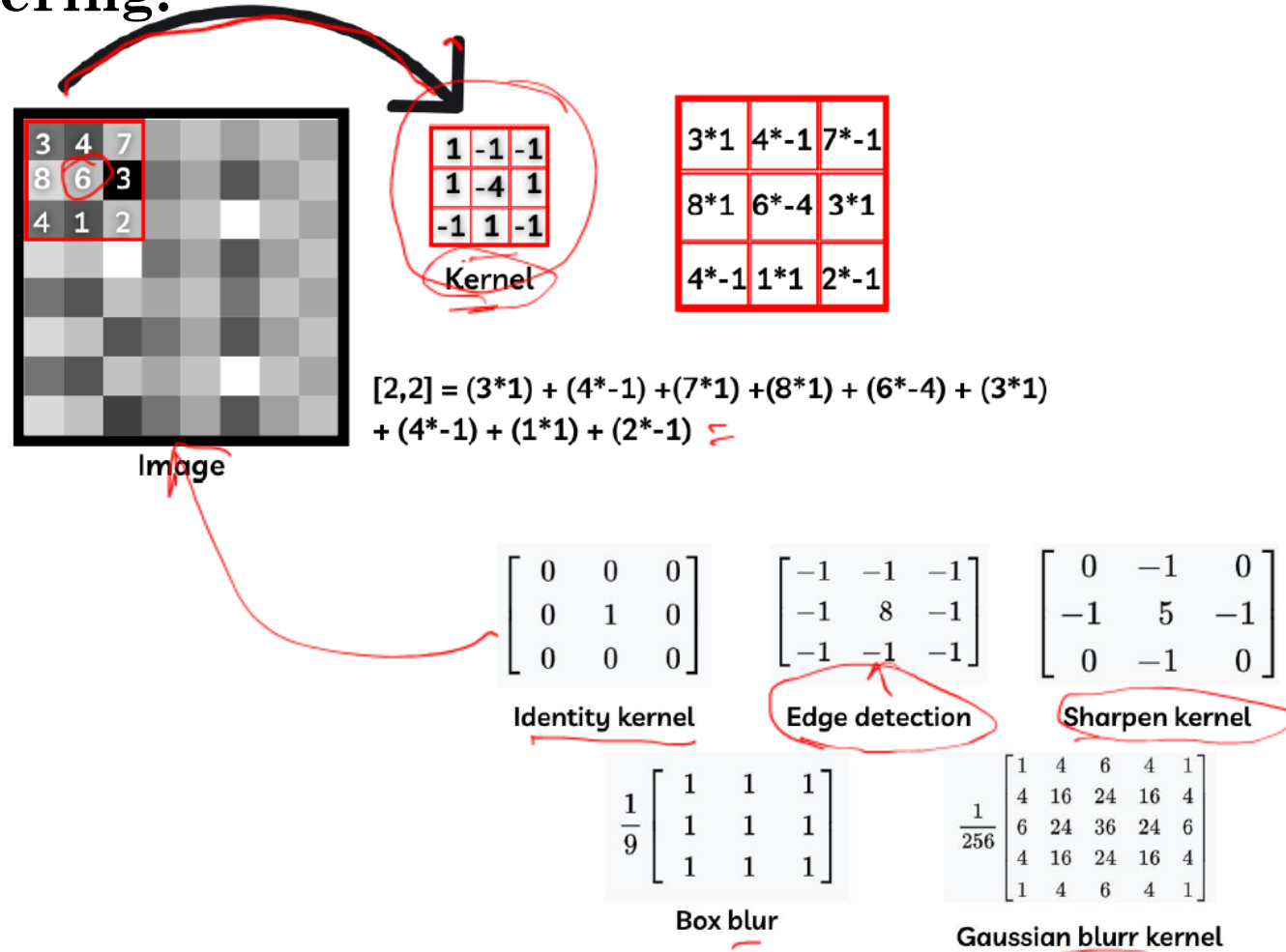


a b

FIGURE 3.32 Two 3×3 smoothing (averaging) filter masks. The constant multiplier in front of each mask is equal to 1 divided by the sum of the values of its coefficients, as is required to compute an average.

Digital Image Processing (DIP)

- Spatial Filtering:



References

- https://www.canerozcan.net/index_en.html

